



School of Computing

M33145-Programming for Data Analysis and AI

Coursework

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1.Descriptive analytics

1.1 Statistics For All Attributes

Table 1.1 Attributes and their datatypes.

Numerical	Categorical
id	gender
age	hypertension
avg_glucose_level	heart_disease
bmi	ever_married
	work_type
	residence_type
	smoking_status
	stroke

	age	avg_glucose_level	bmi
count	5110.000000	5110.000000	4909.000000
mean	43.226614	106.147677	28.893237
std	22.612647	45.283560	7.854067
min	0.080000	55.120000	10.300000
25%	25.000000	77.245000	23.500000
50%	45.000000	91.885000	28.100000
75%	61.000000	114.090000	33.100000
max	82.000000	271.740000	97.600000

Fig 1.1 Numerical attributes and their statistical analysis

Fig1.1 shows that 'age' ranges from 0.08 to 82 years and has an even spread of data. The mean and std of 'avg_glucose_level' suggests there is a wide spread of data too whilst BMI has a large range but most of the data is located around the mean.

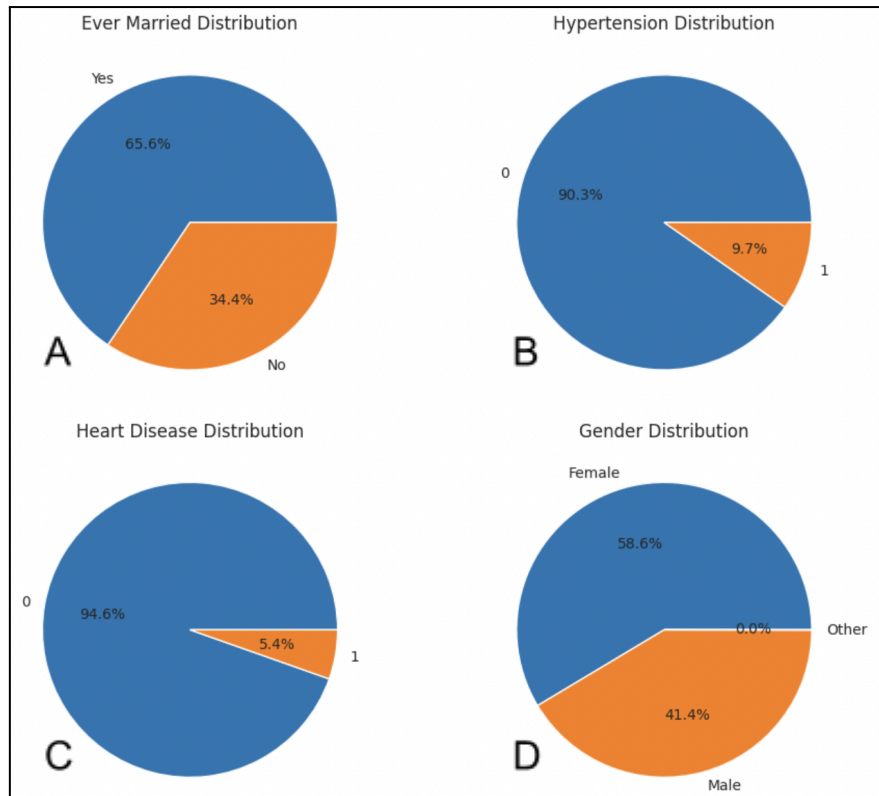


Fig.1.2 Pie charts of categorical attributes. In both Hypertension(B) and Heart Disease(C) Distribution 0 = No and 1 = Yes they have had a stroke.

Fig.1.2 (A&D) presents a relatively even distribution of both genders and if they have ever been married before. However, 'B'&'C' both have <10% being positive for hypertension and heart disease. This small representation may make it hard to see a visible correlation between other attributes.

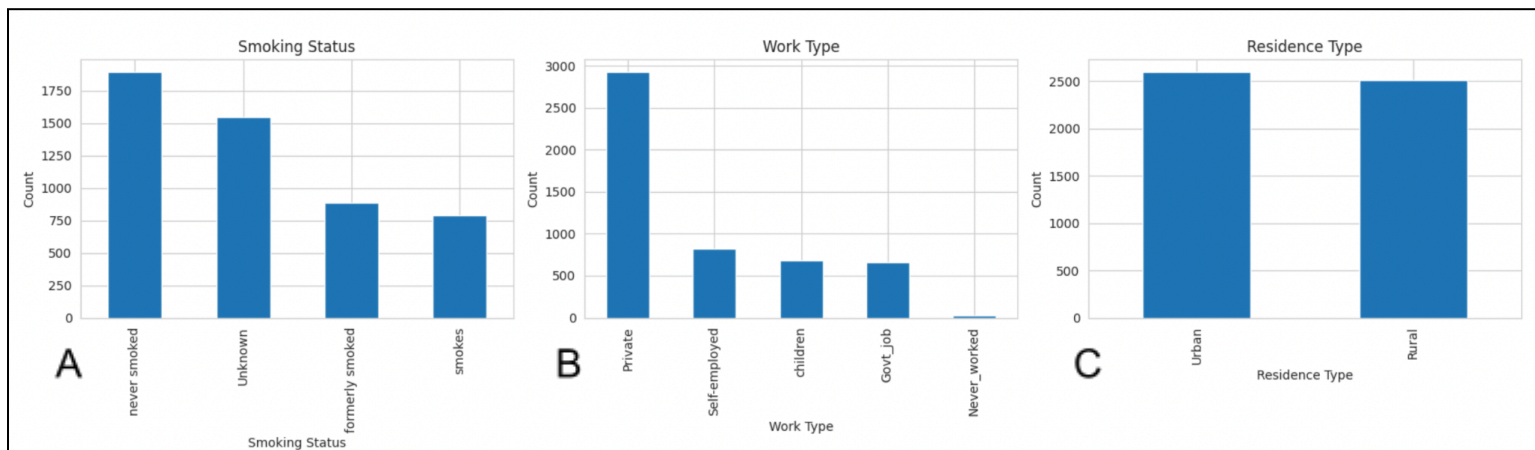


Fig.1.3 Histograms of categorical attributes.

Fig.1.3(A) has good smoking status representation between the groups but there is a large value of 'unknown' data which will be taken into consideration during data preparation. A large proportion of Work Type(B) is 'private' and there are nearly none that have 'never_worked'. Residence type(C) has an even distribution of data within categories.

1.2 Data Visualisation

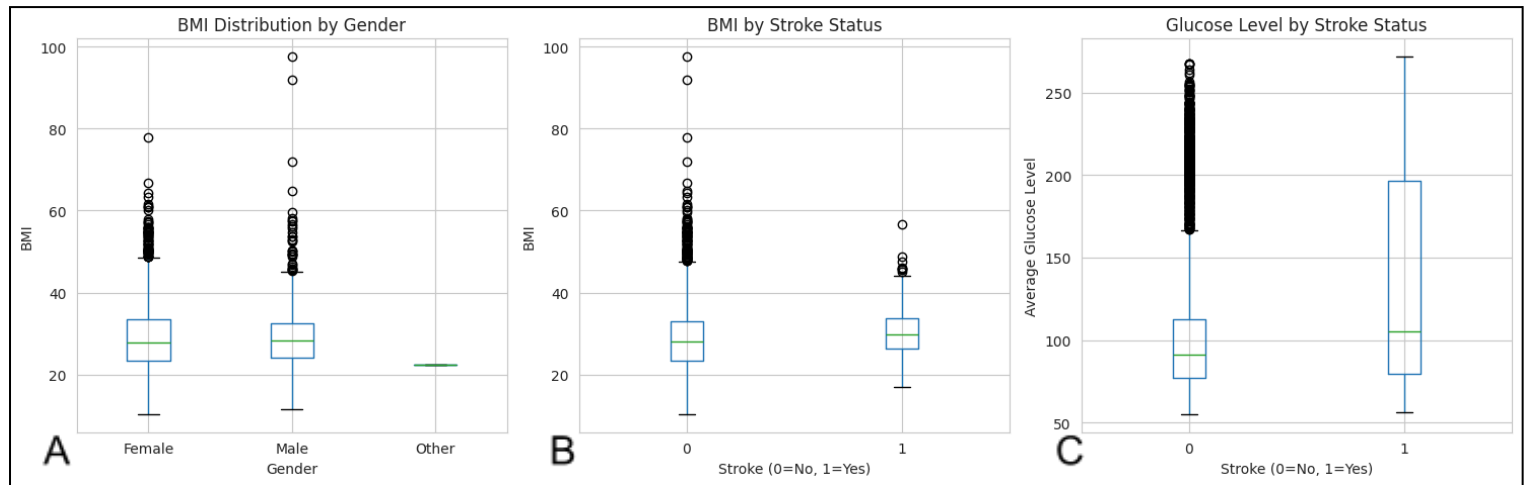


Fig.1.4 Boxplot distribution comparisons. A)BMI against gender. B)BMI against stroke. C)Glucose against stroke.

Table.1.2 observation/comparison/interpretation of Fig.1.4.

	BMI Distribution by Gender	BMI by Stroke Status	Glucose Level by Stroke Status
Key Distributional Observations	All gender groups show right-skewed BMI distributions with numerous high end outliers. Male and female groups have similar IQRs.	Both stroke and non-stroke groups exhibit right skew and extreme outliers. Stroke group has a slightly higher median BMI and a tighter IQR.	Strong right skew in both groups, with higher upper quartile(1=yes) and extreme outliers(0=no). Stroke(1=yes) group shows a substantially wider IQR.
group Differences	Median BMI is almost identical in males and females, though IQRs overlap substantially. The "Other" category is not statistically reliable.	Median BMI is higher among stroke patients, though distributions overlap considerably. Non-stroke group shows greater variability.	Stroke(1=yes) patients have a clearly higher median and upper range of glucose levels compared to non-stroke patients.
Interpretation	BMI reflects similarly across male and female, suggesting gender alone does not strongly differentiate BMI related stroke risk; sample imbalance limits inference for the "Other" group.	Elevated BMI appears associated with stroke, but overlapping suggests BMI alone is insufficient as a discriminative predictor. But it may be a risk factor in combination.	Average glucose level demonstrates a strong association with stroke, indicating hyperglycaemia as a key risk factor and a potentially more powerful predictor than BMI.

Overall, while BMI shows small differences across gender and stroke status, average glucose level exhibits a clinical correlation among stroke patients; a stronger association with stroke risk.

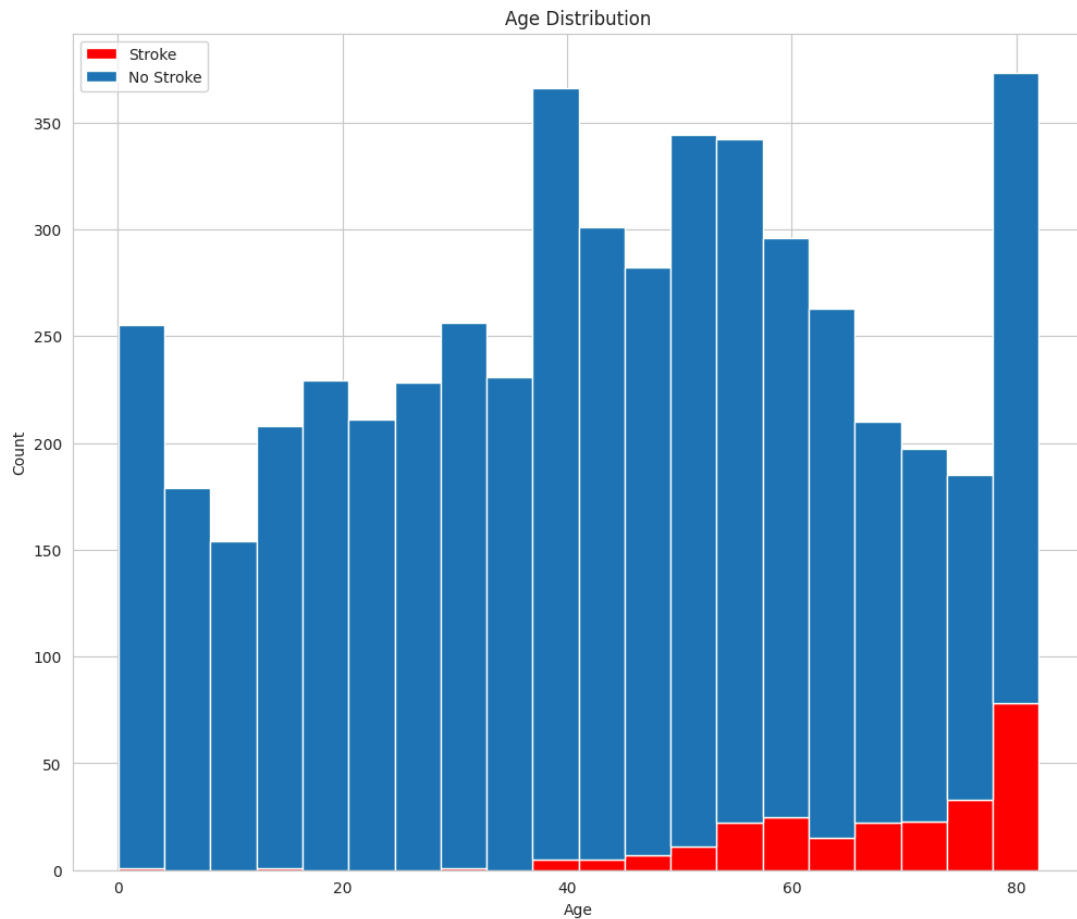


Fig.1.5. A histogram showing the demographic comparison between two attributes, age and stroke/no stroke.

Histogram Fig.1.5 shows a strong positive correlation with stroke incidence(red). Stroke is rare at younger ages and increases from midlife onward, especially after age 50. Non-stroke is a broad population across all ages, however the trend of stroke cases indicates that age is a dominant, non-linear risk factor for stroke.

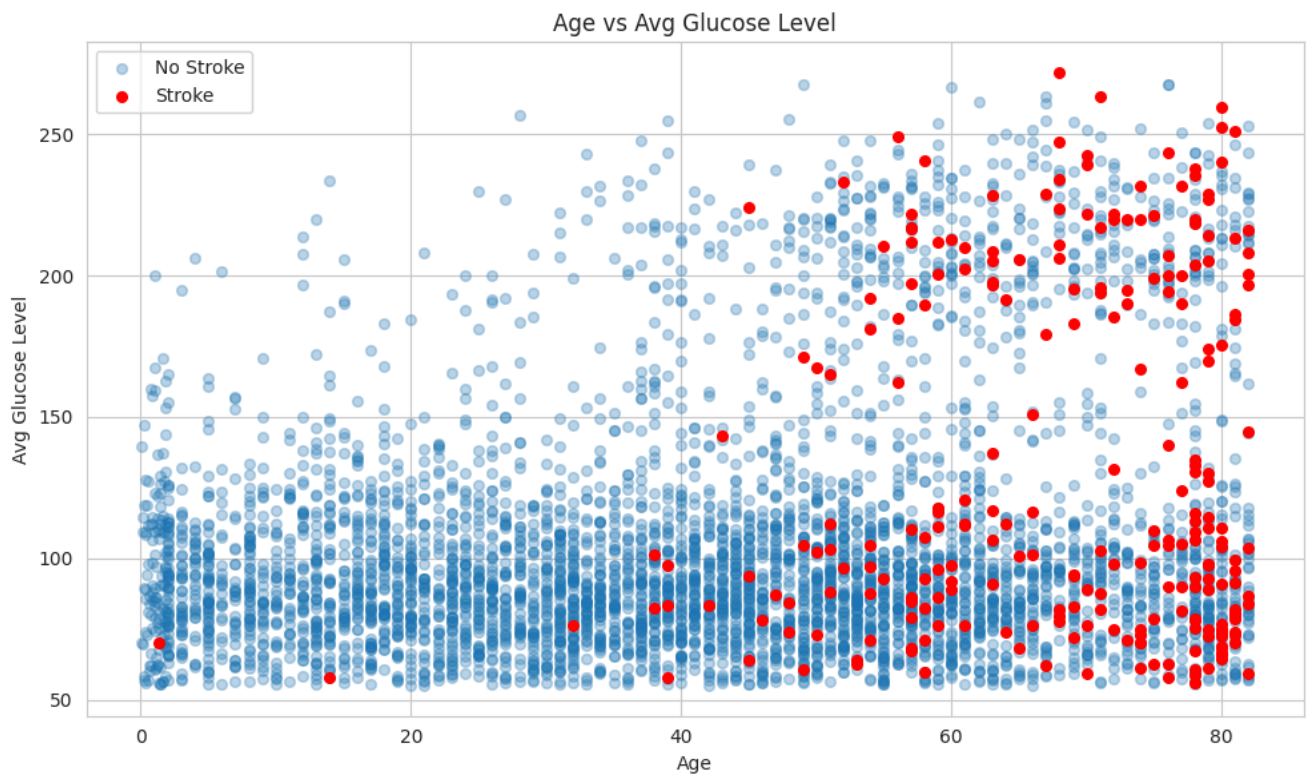


Fig.1.6. A scatter graph highlighting the distribution of 'Avg Glucose Level' against 'Age', and if there was an incidence of a stroke.

This scatter graph presents that stroke cases cluster disproportionately at older ages and higher average glucose levels, thus a clear interaction between age and glycaemic exposure in stroke risk. Exposure of high glucose levels across all ages in the data, with the likelihood of a stroke increasing, suggests a compound risk factor.

Correlation Matrix of All Variables

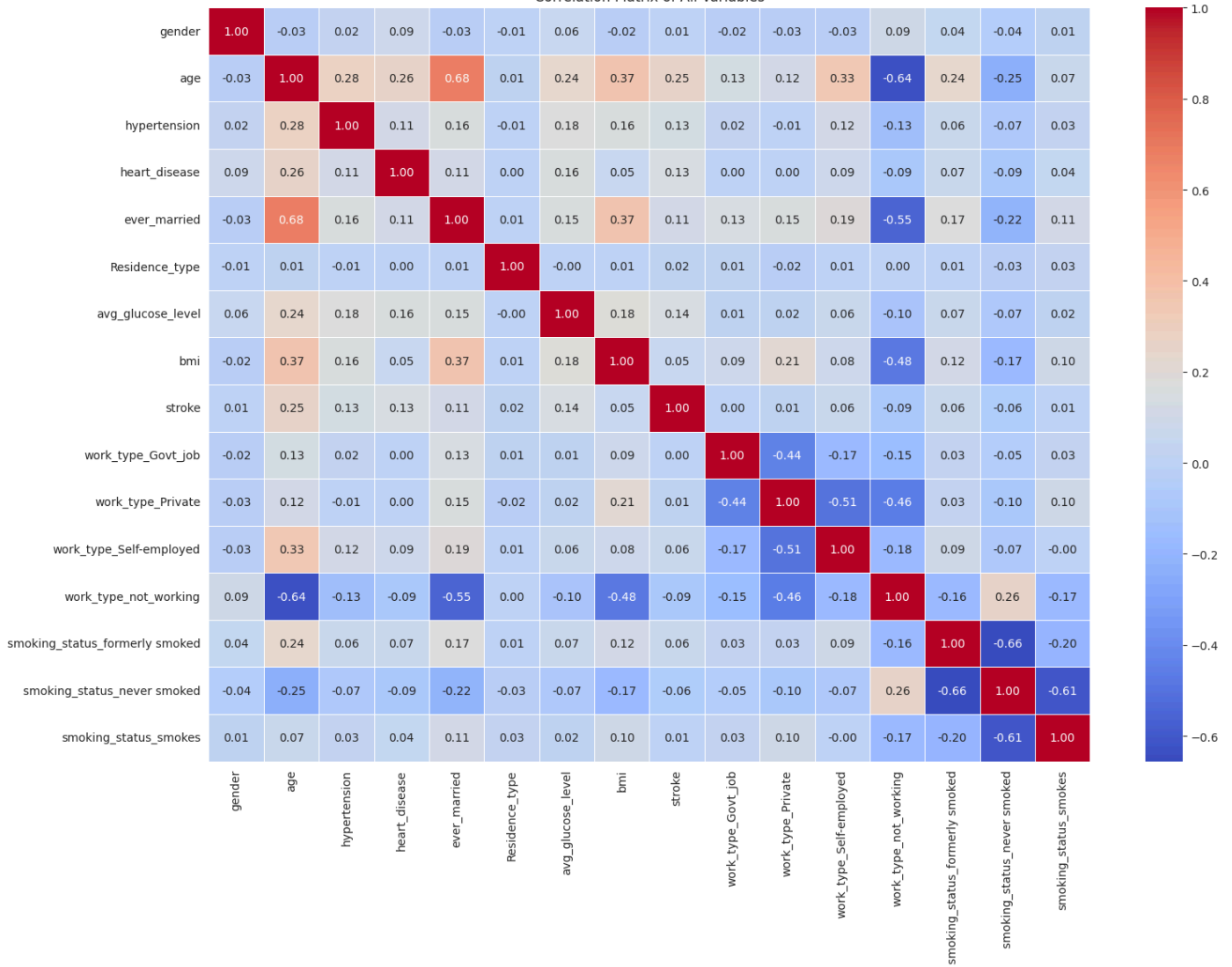


Fig.1.7. A Heatmap of all attributes.

Fig.1.7 indicates that age is moderately associated with stroke, as well as age and average glucose level. BMI, hypertension, and heart disease have weak correlations but are positive. Strong correlations among social demographic variables (e.g. age with marital and work status) suggest multicollinearity, that stroke is multifactorial and unlikely swayed by a single variable.

2.Data preparation

To enable robust analysis, the stroke dataset underwent preprocessing steps to address missing values, rare categories, outliers, and class imbalance across numerical and categorical features. These included data cleaning, missing value imputation, feature engineering, outlier treatment, categorical encoding, and task-specific preparation for classification, regression, and clustering. Collectively, this systematic approach ensures that the dataset is well-prepared for reliable, unbiased, and interpretable model development

2.1 Removal of the ‘Other’ Gender Category

Table.2.1: Gender Category Frequencies Before Cleaning

	Gender	Count	Percentage
0	Female	2994	58.59%
1	Male	2115	41.39%
2	Other	1	0.02%

Initial inspection of the *gender* variable revealed the presence of an “Other” category with only a single observation. Due to its extremely low frequency, this category was removed prior to further analysis. With only one instance, the category cannot support meaningful statistical learning and would result in a near-zero variance dummy variable after one-hot encoding, contributing noise rather than signal. The removal affects approximately 0.02% of the dataset and therefore has a negligible impact on representativeness while simplifying downstream encoding. After removal, the dataset contained 5,109 observations across two categories (Female and Male), ensuring sufficient support for stable parameter estimation.

2.2 Handling Missing Values in Body Mass Index (BMI)

Table.2.2: Median BMI by Age Group and Gender (Used for Imputation)

	age_group	gender	Median_BMI	Non_missing_n
1	0–18	Female	20.40	441
2	0–18	Male	20.00	454
3	19–30	Female	26.50	430
4	19–30	Male	27.00	210
5	31–45	Female	29.30	657
6	31–45	Male	30.65	354
7	46–60	Female	29.80	655
8	46–60	Male	30.90	490
9	>60	Female	29.10	714

10	>60	Male	29.10	503
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BMI contained 201 missing values, requiring careful treatment due to the known association between BMI, age, and gender. Missing BMI values were imputed using the median BMI within each age-group gender combination, calculated from non-missing observations (Table 2.2). For example, females aged 46–60 had a median BMI of 29.8, while males aged 19–30 had a median BMI of 27.0. This stratified median imputation reflects systematic demographic patterns: BMI tends to increase with age, and biological sex influences body composition. Median imputation was preferred to global mean imputation (BMI = 28.89), which would ignore these relationships and introduce bias by assigning identical values across heterogeneous groups. Median is also more robust to extreme values. Following imputation, the BMI variable contained no missing values, preserving realistic distributions while ensuring completeness.

2.3 Handling ‘Unknown’ Values in Smoking Status

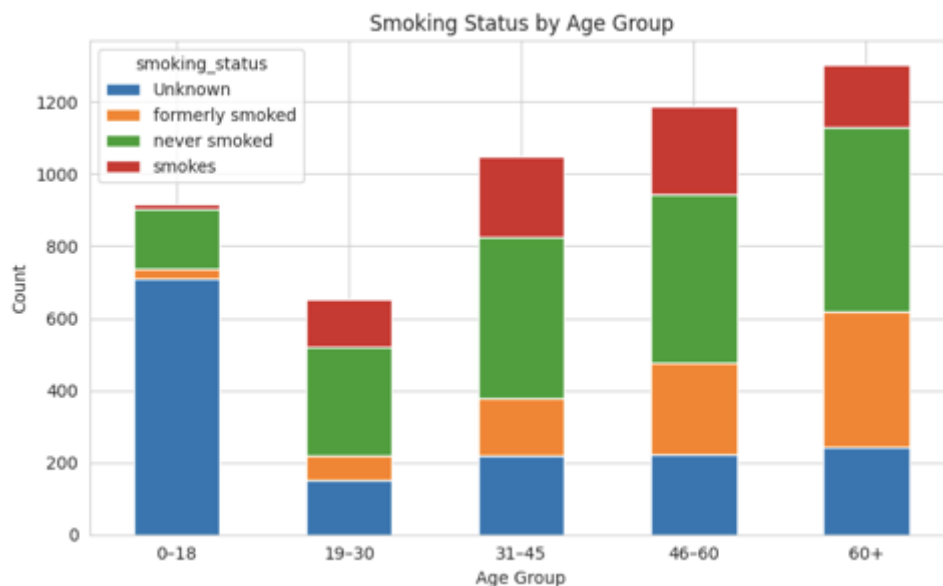


Figure.2.2: Smoking Status Distribution by Age Group (Before Imputation)

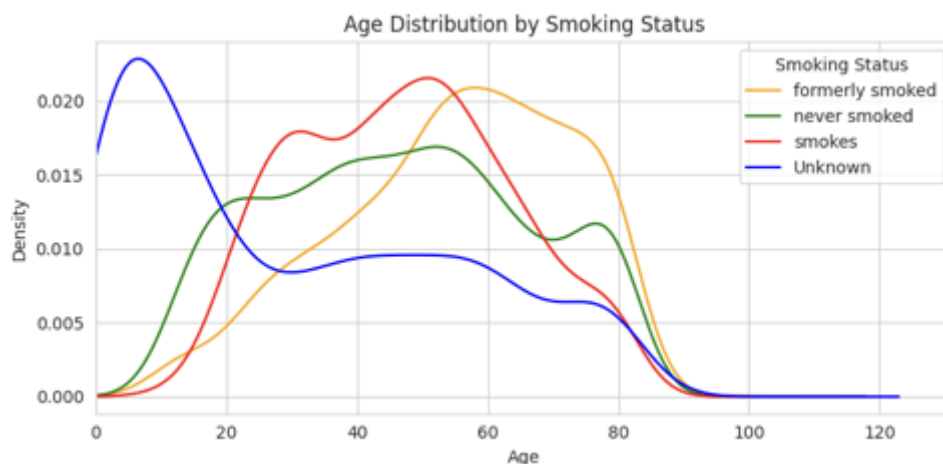


Figure.2.3: Smoking Status Proportions Across Age

'Smoking status' contained an "Unknown" category, which does not represent a meaningful smoking behaviour and could distort model learning if left untreated. Exploratory analysis reveals that the "unknown" category is heavily concentrated in the youngest demographic (ages 0-18), where it represents the clear plurality. This pattern indicates that "Unknown" largely reflects missing or unrecorded information rather than a distinct smoking behaviour. Age-stratified analysis further revealed meaningful trends: individuals aged 20–60 are predominantly current smokers, while the 60+ group is dominated by former smokers. Based on these patterns, *Unknown* values were imputed using the modal smoking category within each age group, rather than applying global mode imputation.

Table.2.3: Smoking Status Distribution Before vs After Age-Stratified Imputation

Smoking Status	Count Before	Percentage Before	Count After	Percentage After (%)
Never smoked	1892	37.03%	3436	67.25%
Smokes	789	15.44%	789	17.30%
Formerly smoked	884	17.30%	884	15.44%
Unknown	1544	30.22%	0	0.00

Applying age-stratified mode imputation prevents artificial inflation that would occur with global imputation. Instead of forcing all 1544 "unknown" cases into one category, this method respects the distribution of smoking across ages: younger individuals are more likely never to have smoked, while older individuals are more likely to be former smokers. Following this adjustment, the "smoking status" variable now depicts a more representative distribution.

2.4 Feature Engineering of Work Type

Table.2.4: Work Type Frequencies Before and After Merging

Work Type	Before Feature Engineering	After Feature Engineering
Private	2924	2924
Self-employed	819	819
Children	687	0
Never_worked	22	0
Govt_job	657	657
Not working	0	709

Analysis of the *work_type* variable showed that the “Never_worked” category contained only 22 observations (less than 0.5% of the dataset). Such a small category would produce sparse dummy variables after one-hot encoding, reducing statistical reliability. To address this, the “Never_worked” category was merged with “Children”, forming a new category labelled “Not_working”. Both groups represent individuals without employment history, making the merge semantically coherent. This step reduced dimensionality, improved category balance, and enhanced the robustness and interpretability of the *work_type* feature for downstream models.

2.5 Handling Extreme Outliers Using Winsorisation

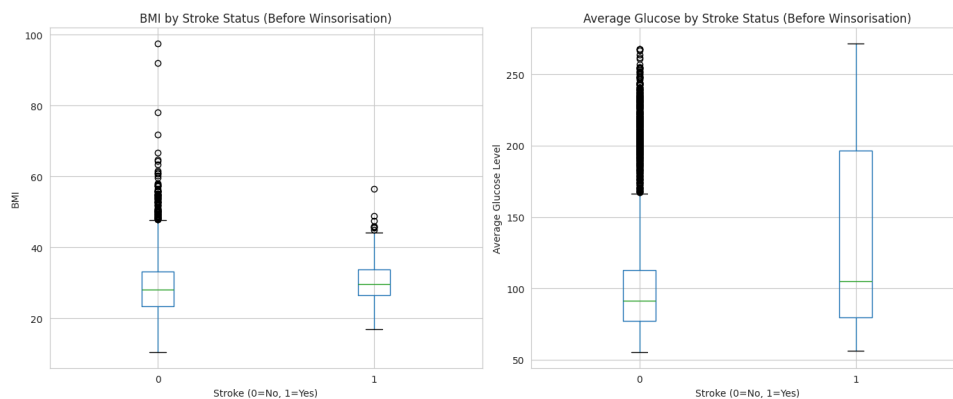


Figure.2.5: Glucose Level and BMI Distribution by Stroke Status (Before Winsorisation)

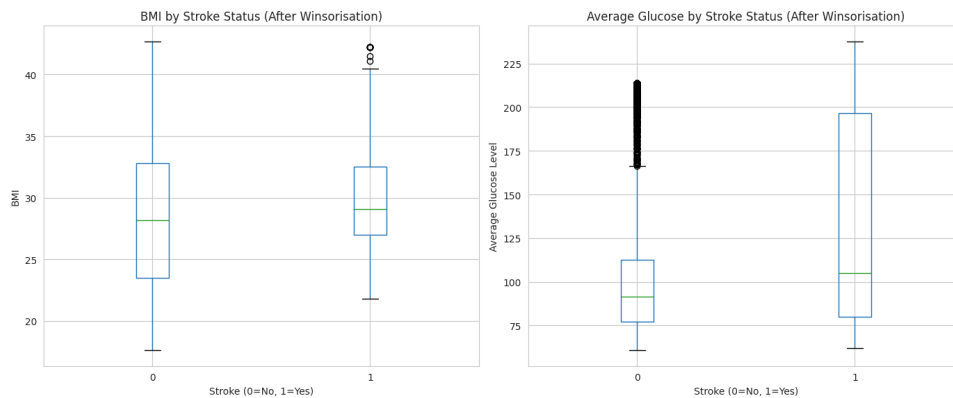


Figure.2.6: Glucose Level and BMI Distribution by Stroke Status (After Winsorisation)

Average glucose levels and BMI exhibited extreme values, particularly within the minority stroke class. For instance, glucose levels ranged from 55.12 to 267.76 mg/dL in the non-stroke group and from 56.11 to 271.74 mg/dL in the stroke group, inflating variance and potentially obscuring meaningful patterns. To mitigate this, class-conditional winsorisation was applied at the 5th and 95th percentiles for both BMI and glucose levels. Values below the 5th percentile were set to the lower cap, and values above the 95th percentile were set to the upper cap. This approach preserves class-specific distributional differences while reducing the influence of extreme outliers, aligning with best practices for univariate outlier handling when retaining all data points is important (Galli, 2022, p. 304).

Table.2.5: Summary Statistics Before and After Winsorisation

Class	Feature	Mean (Before)	Mean (After)	SD (Before)	SD (After)
Non-stroke	Glucose Level	104.79	104.11	43.85	41.38
Stroke	Glucose Level	132.54	132.08	61.92	60.64
Non-stroke	BMI	28.81	28.58	7.80	6.79
Stroke	BMI	30.26	30.14	5.85	5.17

Table 2.5 demonstrates reduced variance with minimal impact on means. Importantly, all observations, including all 249 stroke cases, were retained, maintaining minority-class representation in this highly imbalanced dataset (19.52:1). While winsorisation introduces slight clustering at percentile boundaries, it substantially improves numerical stability for later modelling.

2.6 Removal of Non-Informative Features

Two features were excluded from modelling: the 'id' variable serving solely as a record identifier, and 'age_group' created temporarily for imputation purposes. The "id" variable provides no information about stroke risk, containing only administrative metadata unsuitable for predictive modelling. The "age_group" variable (five categories based on age ranges) was removed to prevent redundancy with the continuous age variable already present. Retaining both introduces multicollinearity as age_group derives deterministically from age, providing no additional information whilst increasing dimensionality.

2.7 Categorical Encoding Strategies

Table.2.6: Encoding Strategy by Variable Type

Variable	Original Type	Encoding Method	Number of Features After Encoding
gender	Nominal	One-hot	2
work_type	Nominal	One-hot	4
residence_type	Nominal	One-hot	2
smoking_status	Nominal	One-hot	3
ever_married	Binary	Label	1
hypertension	Binary	Label	1
heart_disease	Binary	Label	1
stroke	Binary	Label	1

Categorical variables received encoding based on their properties. Binary variables (ever_married, hypertension, heart_disease, stroke) underwent label encoding. These

variables are inherently binary, and label encoding provides a simple and interpretable numeric representation without loss of information. Nominal variables lacking inherent ordering (gender, work type, residence type and smoking status) underwent one-hot encoding, creating binary indicator variables for each category. Gender expanded to two indicators (Female, Male), work_type to four indicators (Government job, Not working, Private, Self-employed), Residence_type to two indicators (Rural, Urban), and smoking_status to three indicators (formerly smoked, never smoked, smokes). This transformation increases dimensionality from 4 to 11 features whilst avoiding false ordinality that would arise from assigning arbitrary integer codes (0,1,2,3) to unordered categories. One-hot encoding ensures model coefficients reflect true category effects rather than artefacts of arbitrary numeric assignments.

2.8 Classification PreparationTable

Table 2.7: Class Distribution Across Dataset Splits

Split	Non-stroke	Stroke	Ratio
Original	4,860	249	19.52:1
Train (before SMOTE)	3888	199	19.54:1
Train (after SMOTE)	3888	3888	1:1
Test	972	50	19.44:1

The dataset uses a 80/20 split into training and test sets using stratified sampling to preserve the 19.52:1 class imbalance ratio. Training data contained 3888 non-stroke and 199 stroke cases, whilst testing contained 972 non-stroke and 50 stroke cases. Stratification preserves the original class distribution across splits, ensuring fair evaluation and reliable model tuning. To address the severe class imbalance in the training data, SMOTE was exclusively applied, generating synthetic stroke cases. The balanced training set (7,776 records: 3888 per class) enables models to learn minority patterns without overwhelming majority class dominance. Test data remained unchanged at the original imbalanced distribution (972:50 ratio), ensuring evaluation reflects realistic deployment conditions where class imbalance persists. StandardScaler was fitted on the training data and subsequently applied to the test sets. Feature scaling is required for distance-based and gradient-based algorithms, ensuring that all features contribute equally to model training.

2.9 Regression Preparation

For regression tasks involving continuous target variables, a similar but simplified preparation process was applied. The data was split into training and test sets using the same proportions as classification. StandardScaler was fitted on the training set and applied to the remaining splits. Scaled datasets were produced for regression analysis.

2.10 Clustering Preparation

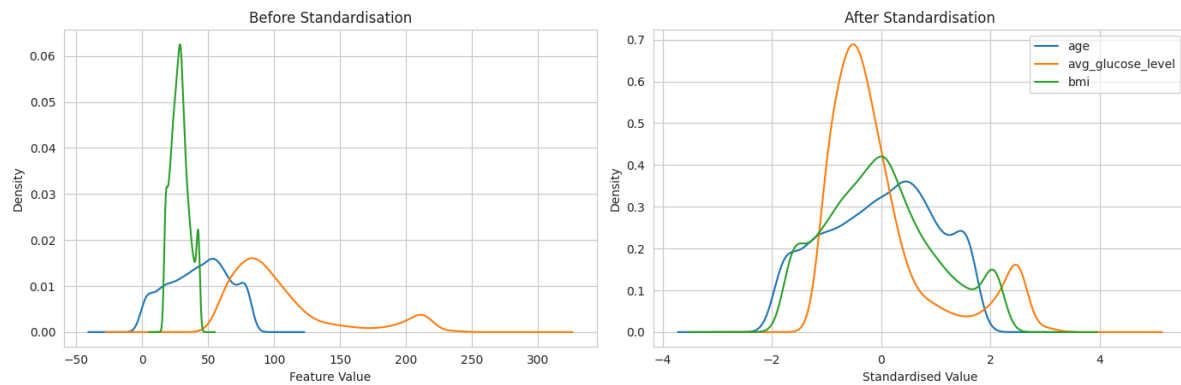


Figure.2.7: Feature Distributions Before and After Standardisation

Clustering represents an unsupervised learning task and therefore requires a distinct preparation strategy. The target variable (stroke) was removed prior to clustering. All numerical features were standardised using StandardScaler to equalise feature scales, which is critical for distance-based algorithms such as K-Means. Standardisation ensured fair distance computation across features originally spanning disparate ranges (e.g., age: 0–82 years; glucose: 55–272 mg/dL; BMI: 9–48). The resulting scaled feature space enabled meaningful cluster formation driven by structure rather than scale artefacts

3. Classification

Three classification models were implemented to predict stroke outcomes in this dataset: SVM, Random Forest (RF) and XGBoost. The dataset had severe class imbalance with only 249 stroke cases (4.9%) among a total of 5110. An 80-20 stratified split and SMOTE was applied to balance the data. Models were then evaluated on unchanged test data to reflect a realistic performance. Model parameters shown in Table.3.0

Table.3.0 A table showing classification model hyperparameters and rationale.

Model	Key Parameters	Rationale
SVM	kernel='rbf' C=1.0 gamma='scale' class_weight='balanced'	RBF kernel handles non-linear relationships between features and default parameters set for baseline performance
Random Forest	n_estimators=150 max_depth=12 min_samples_leaf=6 class_weight='balanced'	Bootstrap aggregation reduces overfitting. Provides interpretable feature importance rankings critical for clinical contexts.
XGBoost	n_estimators=100 learning_rate=0.1 max_depth=6 subsample=0.8 colsample_bytree=0.8	Sequential tree construction corrects previous errors. Subsampling parameters provide regularization against overfitting.

3.1 Results and Analysis

Table.3.1. A table showing classification report metrics for each model.

Model	Accuracy	Precision	Recall	F1-Score	AUC
SVM	0.92	0.23	0.26	0.25	0.76
Random Forest	0.94	0.38	0.32	0.35	0.81
XGBoost	0.94	0.34	0.34	0.34	0.85

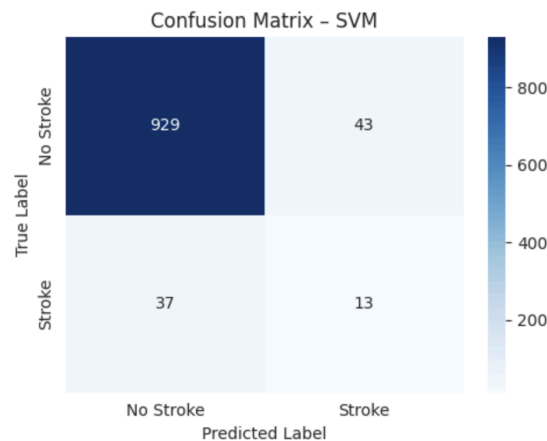


Fig.3.1. SVM Confusion Matrix.

SVM produced the weakest results, catching only 13/50 stroke cases (26% recall), while missing 37 (74% false negative rate). A 23% precision and an AUC of 0.76, the model struggled despite being balanced with SMOTE. The confusion matrix in figure 3.2 shows us 43 false positives (predicted stroke, but no stroke for patient) and 37 false negatives (missed stroke cases), this indicates that SVM defaulted to predicting "no stroke" most of the time. Despite achieving 92% accuracy, the RBF kernel failed to find effective patterns in the 15-dimensional feature space, achieving high accuracy on non-stroke cases at the expense of detecting strokes.

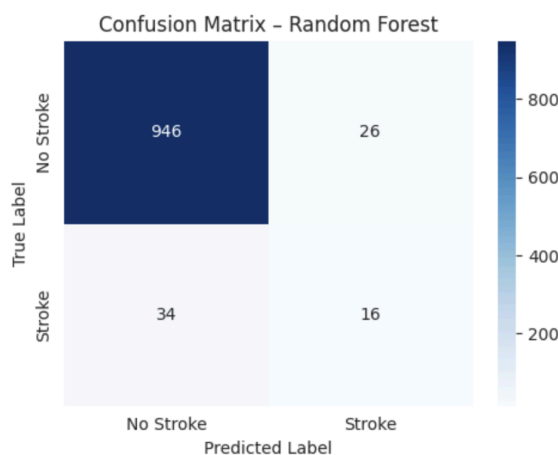


Fig.3.2 - Random Forest Confusion Matrix.

RF performed moderately better, it achieved 32% recall where it detected 16 out of 50 stroke cases and a 38% precision. This improved precision shows us there were fewer false

positives (26) compared to SVM. RF did better than SVM by detecting fewer false negatives, however having 34 false negatives is still not great when detecting strokes. The AUC of 0.81 (Table.3.1) demonstrates that random forest separates stroke from non-stroke cases better than SVM (AUC=0.76). This reflects that RF has superior handling of the features through its ensemble averaging. The F1-Score of random forest (0.35) was also higher than SVM (0.25) which shows us that random forest has a better balance of precision and recall than SVM.

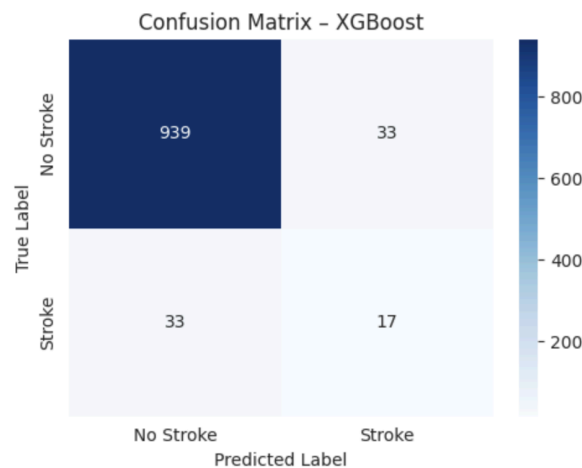


Fig.3.3. XGBoost Confusion Matrix.

XGBoost delivered the strongest performance with 34% recall, where it correctly identified 17/50 stroke cases and missing 33. Both precision and recall was the same at 34% showing a balanced error distribution which may suggest consistent classification behaviour. RF achieved a slightly higher precision (0.38), however it is more important in a healthcare setting for the recall to be higher to minimise the amount of missed stroke cases. XGBoost achieved the highest AUC (0.85) out of all models which demonstrates that it was better at separating stroke from non-stroke cases. This is likely due to the advantage XGBoost has over random forest where it is sequentially learning from each tree and fixing mistakes from previous trees.

3.2 Feature Importance

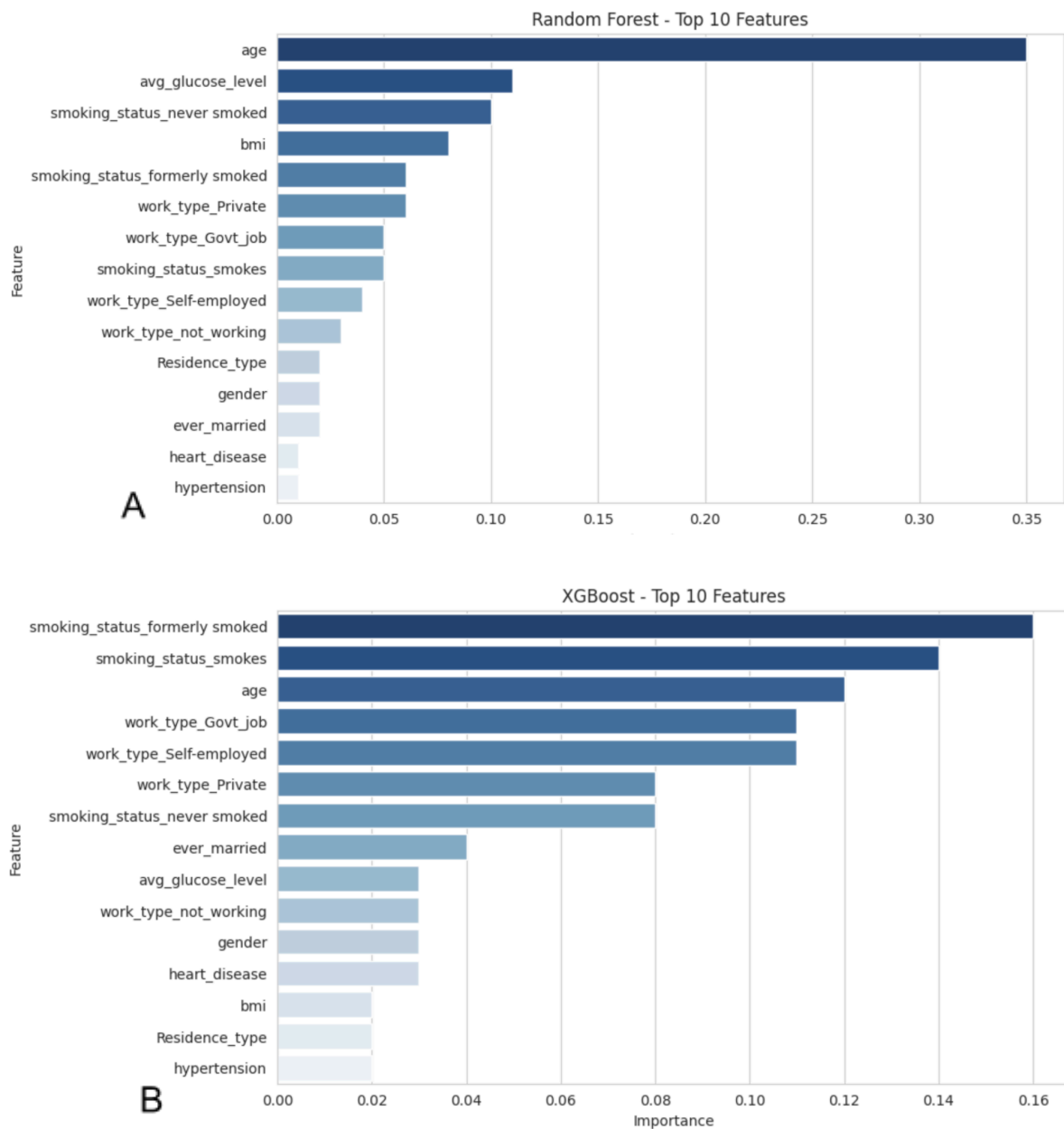


Fig.3.4. Top 10 features of Random Forest and XGBoost

Feature importance is shown in Fig.3.4, it reveals contrasting results between the tree based models. RF identified age as the most dominant predictor (35% importance), followed by average glucose levels (11%) and a smoking status of never smoked (10%). The drastic drop of importance from age to other features suggest that RF is heavily reliant on age when making its classification decisions, while the other features had minor importance.

However, we can see that XGBoost has distributed importance more evenly, with smoking status formerly smoked (16%), currently smoked (14%) and age (12%) being top three most important features with others that are close to them. This heavily contrasts with the random forest model. The differences reflect how the two models learn from the data. RF builds trees independently allowing more dominant features such as age to repeatedly drive early splits

whereas XGBoost builds trees sequentially and focuses on correcting previous errors. And it shows us how XGBoost has more of an even feature importance across the features.

Another thing to note is that the models differ in how they rank smoking. RF emphasises never smoked (10%), while XGBoost places a greater emphasis on formerly smoked (16%). This suggests that each model is capturing different patterns in the data. RF appears to treat never smoking as a general distinguishing factor, whereas XGBoost, through its step-by-step learning process, gives more weight to former smoking when refining predictions after stronger features such as age have already been considered. Both models agreed that age and smoking-related features to be important stroke predictors which makes sense given the general consensus on stroke risk factors.

3.3 Comparative Analysis

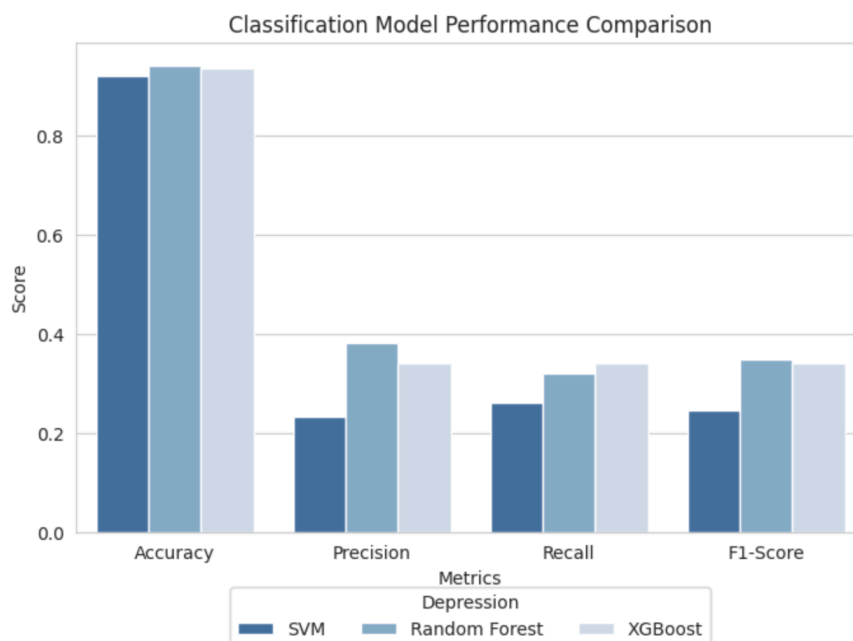


Fig.3.5. Grouped Bar Chart of Classification Model Performance

Critical findings show us poor metrics all across the board (Fig.3.5) apart from each model having a really high accuracy. This suggests to us that none of the classifying models achieved an acceptable level of stroke detection. In a healthcare dataset where we are predicting stroke, catching fewer false negatives is paramount as we want to avoid any of them in a healthcare setting as it can pose serious consequences. XGBoost gave us the best model in terms of detecting the fewest false negatives but on the whole all models would most likely be unsuitable for a clinical environment.

3.4 Limitations and Conclusion

Several limitations affect model performance. Even though data was balanced using SMOTE to create synthetic stroke cases, it may not have fully captured real stroke presentation. This may have caused overfitting to artificial patterns and failed to generalise from the data. The consistent low recall (26-34%) across each model was likely due to the dataset being so imbalanced rather than algorithmic failure, with only 4.9% of the total dataset attributing to the target value of stroke, it seemed to be too insufficient for robust learning. Using these models in a clinical setting is inadvisable due to the high error rate of detecting actual strokes, in a medical setting you want to minimise the amount of false negatives you get, XGBoost did the best with this as it had the lowest false negative rate (33) but it would not be safe to use in a clinical environment.

Concluding, XGBoost achieved the best performance with 34% recall (17/50 stroke cases detected and 33 missed cases of stroke), 34% precision and an AUC of 0.85. followed by Random Forest (recall=32%, AUC=0.81) and SVM (recall=26%, AUC=0.76). Feature importance showed us that age and smoking were primary stroke predictors and also revealed algorithmic specific ranking of patterns as well. It would be really difficult to use these models in a clinical setting due to the high error rate, in future having more comprehensive features such as family history, detailed medical records etc with a larger sample of stroke cases in the dataset would most likely optimise models better.

4. Regression

4.1 Regression Tree Regressor

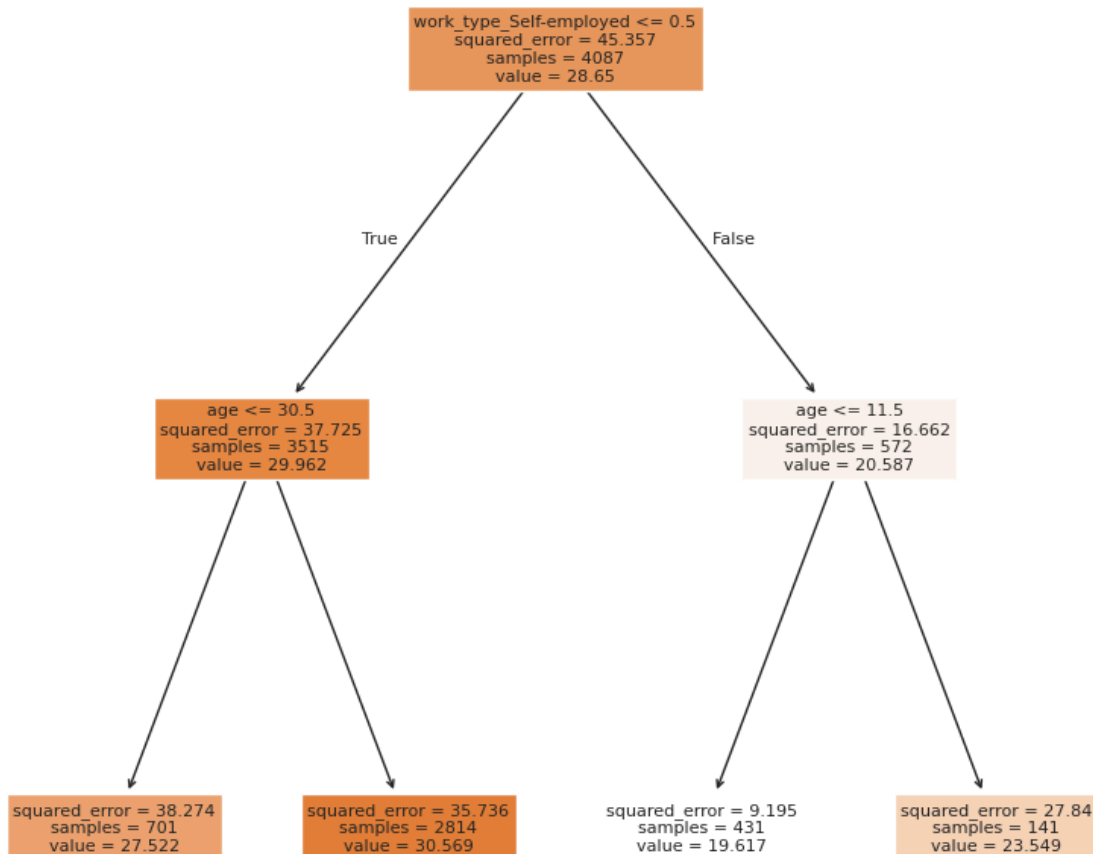


Fig.4.1 Decision Tree Regressor on BMI.

In Fig.4.1 the root node indicates that self-employment status is the strongest determinant of the target variable BMI. This initial split suggests that occupational context, likely reflecting lifestyle structure and work autonomy are greatly influential in regards to BMI variation.

In the true branch (Self-employed ≤ 0.5), those who are not self-employed, the tree further splits on age ≤ 30.5 , indicating that age is a key stratifying factor when occupational lifestyle is more structured. Younger individuals in this group exhibit a lower predicted BMI, while those above 30.5 years show a higher mean BMI, reflecting age-related metabolic and lifestyle changes.

In the false branch (Self-employed > 0.5), the model splits on age ≤ 11.5 , capturing a subgroup likely corresponding to children within self-employed households. This split indicates that age dominates BMI prediction in this subgroup, with older individuals having substantially higher predicted BMI values than younger individuals.

The root node's squared error (45.357) indicates moderate variance in BMI across the full dataset, which consistently decreases down the tree, illustrating variance reduction. Leaf

node errors range from 9.195 (young self-employed subgroup) to 38.274 (younger not self-employed adults), with the lowest error corresponding to a smaller, more homogeneous subset of the population.

Sample distribution across branches shows that the model prioritises not self-employed adults, which constitute the majority of observations, while self-employed subgroups, though smaller, yield more precise BMI predictions due to reduced heterogeneity.

Overall, the tree is shallow and highly interpretable, suggesting low variance and reduced risk of overfitting; however, this simplicity introduces potential bias and underfitting, as BMI is influenced by complex, non-linear interactions (e.g., diet, physical activity, and socioeconomic status) not fully captured by age and employment type alone.

4.2 Linear Regression

OLS Regression Results						
Dep. Variable:	bmi	R-squared:	0.264			
Model:	OLS	Adj. R-squared:	0.262			
Method:	Least Squares	F-statistic:	140.7			
Date:	Wed, 31 Dec 2025	Prob (F-statistic):	0.00			
Time:	15:35:25	Log-Likelihood:	-16202.			
No. Observations:	5109	AIC:	3.243e+04			
Df Residuals:	5095	BIC:	3.252e+04			
Df Model:	13					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	15.6313	0.200	78.311	0.000	15.240	16.023
gender	0.1913	0.166	1.152	0.249	-0.134	0.517
age	-0.0034	0.006	-0.551	0.582	-0.015	0.009
hypertension	1.7226	0.287	6.011	0.000	1.161	2.284
heart_disease	-0.6499	0.378	-1.721	0.085	-1.390	0.091
ever_married	1.8820	0.239	7.878	0.000	1.414	2.350
Residence_type	0.0900	0.162	0.557	0.578	-0.227	0.407
avg_glucose_level	0.0178	0.002	9.016	0.000	0.014	0.022
stroke	-0.5809	0.392	-1.481	0.139	-1.350	0.188
work_type_Govt_job	5.9542	0.211	28.179	0.000	5.540	6.368
work_type_Private	5.8693	0.133	44.169	0.000	5.609	6.130
work_type_Self-employed	5.4847	0.217	25.254	0.000	5.059	5.910
work_type_not_working	-1.6769	0.244	-6.871	0.000	-2.155	-1.198
smoking_status_formerly smoked	5.4551	0.177	30.757	0.000	5.107	5.803
smoking_status_never smoked	4.9109	0.119	41.334	0.000	4.678	5.144
smoking_status_smokes	5.2654	0.174	30.252	0.000	4.924	5.607
Omnibus:	247.913	Durbin-Watson:	2.012			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	284.885			
Skew:	0.578	Prob(JB):	1.37e-62			
Kurtosis:	2.944	Cond. No.	1.29e+18			

Fig.4.2 Ordinary Least Squares (OLS) regression analysis

With 'bmi' being the dependent variable and $R^2 = 0.264$ plus Adj. $R^2 = 0.262$ this gives a modest 26% variance in age, suggesting other unobserved factors influence age. F-statistic = 140.7, $p < 0.001$ indicates this model is statistically significant and at least 1 independent variable follows a linear relationship.

Table.4.1 key predictors from OLS(fig.4.2)

Attributes	coef	p> t	prediction
work_type_...	Private = 5.87 Govt_job =5.95	< 0.001	Strong positive association. Employment captures lifestyle structure, physical activity, income stability, and stress, strong indicators of bmi

	Self-employed= 5.48		
smoking_status_...	formerly = 5.46 never = 4.91 Smokes = 5.27	< 0.001	Strong positive association. Smoking status appears to influence long-term behavioural patterns affecting bmi.
ever_married	1.88	< 0.001	Strong positive association. Marriage can be associated with lifestyle change.
hypertension	1.72	< 0.001	Strong positive association. Hypertension is strongly related to weight therefore related to bim.
avg_glucose_level	0.018	< 0.001	Strong positive association. High glucose levels are linked to higher bmi.

The Omnibus and Jarque-Bera test statistics ($p < 0.001$) indicate rejection of the null hypothesis of normally distributed residuals, suggesting departures from normality that may arise from model misspecification, the presence of outliers or heavy-tailed errors, and potential heteroskedasticity. The Durbin-Watson statistic (≈ 2.01) indicates no evidence of serial autocorrelation in the residuals. Although residuals exhibit mild positive skewness, the kurtosis is close to that of a normal distribution, implying only moderate deviation from normality.

4.3 Model Comparison

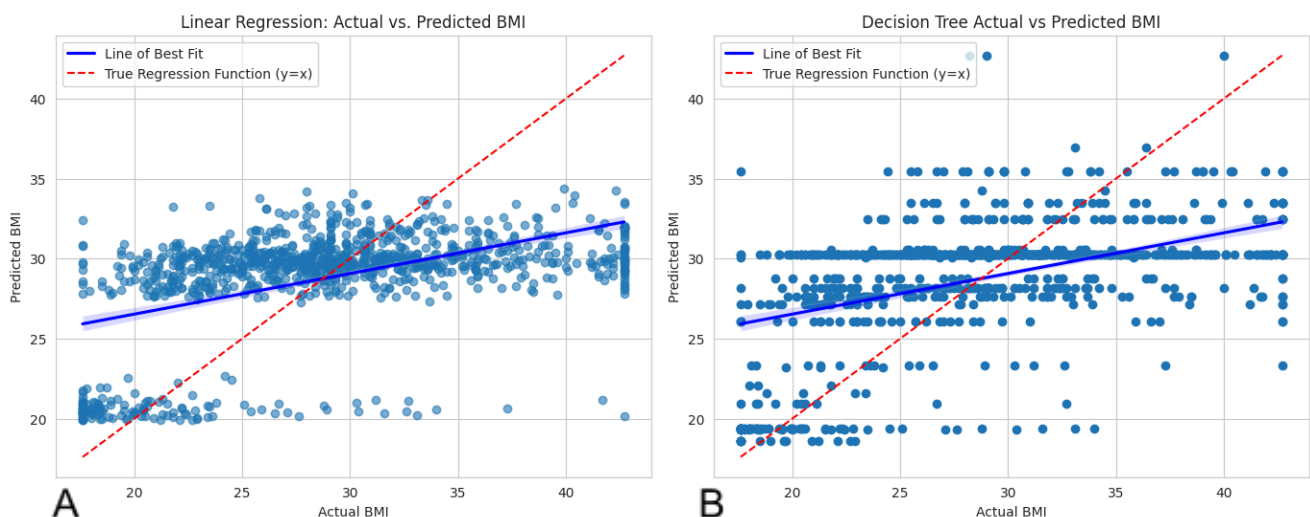


Fig.4.3 Scatter Graphs comparing the actual vs predicted 'bmi' outcomes from regression models. A)Linear Regression outcomes. B)Decision Tree Regressor outcomes.

Both plots in Fig. 4.3 has clustering of predictions around the mid-range of BMI values, particularly between approximately 25 and 35, where points lie closest to the line of best fit and the interception of line $y = x$. For both models, prediction accuracy deteriorates at the extremes of BMI, with systematic underestimation of higher BMI values and over-smoothing of lower values.

The decision tree model (Fig.4.3B) presents a weaker alignment with the true regression line, reflecting its tendency to predict within discrete value bands and regress toward the mean of the training data. This behaviour is characteristic of tree-based models, which struggle with extrapolation beyond observed partitions and produce piecewise-constant predictions.

In contrast, the linear regression model (Fig.4.3A) demonstrates a stronger overall correspondence between actual and predicted BMI, capturing the global linear trend more effectively. However, the dispersion of residuals around the fitted line indicates underfitting, suggesting that linear regression fails to fully capture non-linear relationships present in the data.

Table.4.2 Comparison of predictive performance metrics for Decision Tree Regressor and Linear Regression models.

	Decision Tree	Linear Regression
Mean Absolute Error (MAE)	4.4591424443184655	4.576560912957183
Mean Squared Error (MSE)	32.35408712733139	33.598010080402155
Root Mean Squared Error (RMSE)	5.688065323757401	5.7963790490617635
R2 score	0.2747385945643457	0.24685434903928316
Adjusted R2 score	0.26392455770397316	0.23562454311044545

It is shown in Table.4.2 that the Decision Tree demonstrates slightly better predictive performance for BMI when considering MAE, MSE, and RMSE, having marginally lower average and squared prediction errors compared to linear regression. The regression tree also achieves higher R^2 and adjusted R^2 values, suggesting it explains a greater proportion of variance in BMI and generalises slightly better given the number of predictors. However, the overall low R^2 values (both < 0.3) indicate that BMI is influenced by additional factors not captured in the dataset, limiting the explanatory power of both models.

5. Clustering

5.1 KMeans Clustering

```
# Compute the silhouette score
kmeans_silhouette = silhouette_score(X_cluster_scaled_df, kmeans.labels_).round(2)
kmeans_silhouette

np.float64(0.14)
```

Fig.5.1 silhouette score for KMeans clustering.

KMeans has a Silhouette score of 0.14, while this indicates that the data has been correctly associated to each cluster, there is a large amount of overlap in the clusters.

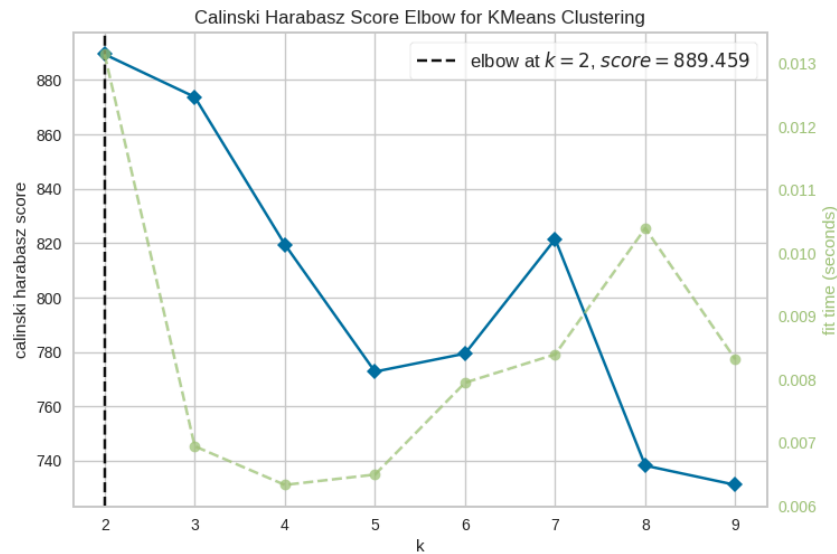


Fig.5.2 Calinski Harabasz Score Elbow for KMeans clustering.

The Calinski Harabasz score suggests that 9 clusters might be better defined in terms of variance ratio. But when $K=2$ is the highest score.

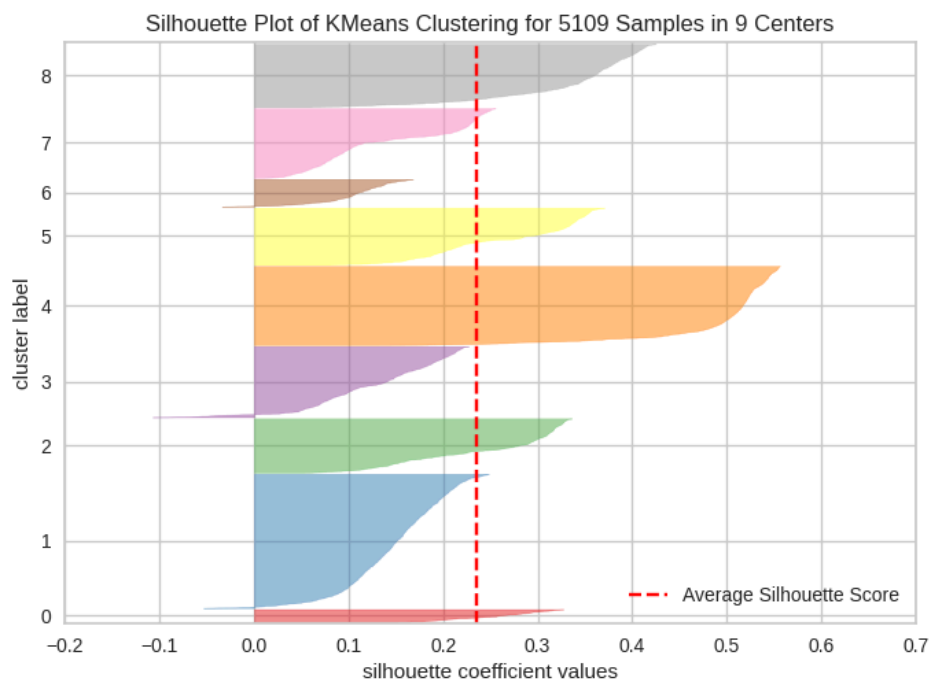


Fig.5.3 Silhouette Plot for KMeans clustering.

Most clusters have a positive silhouette score however most are near to zero with some value below. The average score is ≈ 0.23 indicating moderate clustering quality, but this implies weak separation and could benefit from fewer clusters.

5.2 K-Prototypes Clustering

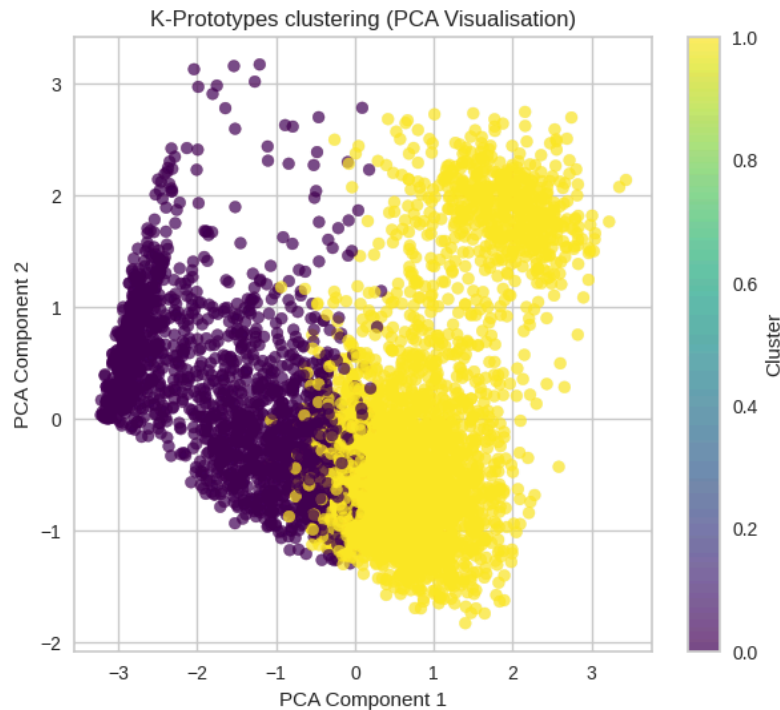


Fig. 5.4 PCA visualisation for K-Prototypes clustering.

The plot on Fig.5.1 shows two different coloured clusters. While there is visible separation just below zero along the 'PCA component 1' x-axis, the clusters overlap slightly along that vertical plane when $x \approx 0.5$ and $y < 1$. This highlights moderate cluster distinctiveness.

As K-Prototypes are designed for mixed data, the separation is likely due to the differences in categorical attributes combined with numeric features. PCA compresses this complexity into two components which may obscure some structure.

The slight overlap indicates that the dataset contains heterogeneous groups with shared characteristics, making perfect segmentation difficult. This could indicate: high intra-cluster variability, possible need for more clusters ($k > 2$) or alternative distance metrics.

5.3 HDBSCAN Clustering

HDBSCAN clustering is a powerful, non-parametric technique that constructs a cluster hierarchy to identify structures of varying density. Unlike k-means, it automatically discovers the optimal number of clusters and designates low-density regions as noise (outliers). Its strength lies in its ability to extract the most stable and prominent clusters from the hierarchy, ensuring robustness against fluctuating local densities and removing the need for a single, global density threshold (E). Therefore the model is well-suited for high-dimensional/complex datasets where cluster definitions are not uniform.

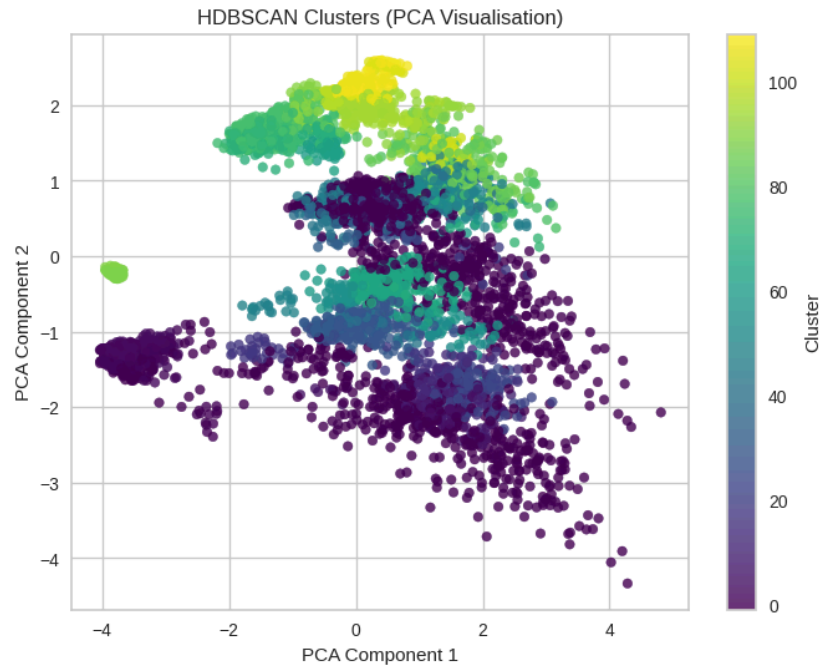


Fig. 5.5 PCA visualisation for HDBSCAN clustering.

The HDBSCAN clustering results show the identification of many small, irregularly shaped clusters, reflecting its density-based nature. Unlike partition-based methods, HDBSCAN allows for variable cluster sizes and implicitly treats low-density regions as noise, although in this projection most points are assigned to clusters.

The clusters in Fig.5.5 appear relatively distinct and packed (excluding the noise) and there are clear spaces between. The slight overlapping indicates relatively good separation. The clusters are densely populated in the center suggesting weak global separability in the data. The visual overlap also highlights a limitation of PCA, as dimensionality reduction may compress or obscure meaningful high-dimensional density structures.

Overall, the large number of clusters and their spatial overlap imply that the data is complex and high-dimensional, with subtle, locally defined patterns that are better suited to density-based clustering approaches than global partitioning methods such as KMeans.

5.4 Clustering Algorithm Comparison

Table.5.1 Davies-Bouldin Index for each clustering algorithm.

Algorithm	Davies-Bouldin Index
K-Prototypes	1.31
HDBSCAN	0.70

The output of HDBSCAN often results in more naturally formed clusters with potentially better-defined boundaries compared to K-Means, and the visualisation helps confirm this by showing distinct, dense regions for each cluster, along with scattered noise points. The

Davies-Bouldin Index (shown in Table.5.2) for HDBSCAN (excluding noise) was reported as 0.70, which is lower than K-Prototypes' 1.31. A lower Davies-Bouldin Index indicates better clustering, conveying that HDBSCAN found more separated and compact clusters.

Ultimately, when looking at the Davies-Bouldin Index in Table.5.1 and the Silhouette score in Fig.5.1, it is devised that KMeans is the least defined clustering model by a huge margin, then K-Prototypes followed, but the HDBSCAN clustering model was indicated that it had the best clusters. When comparing Fig.5.4 and Fig.5.5 the separation of the clusters looks arguably slightly better.

References

Galli, S. (2022). *Python feature engineering cookbook: Over 70 recipes for creating, engineering, and transforming features to build machine learning models*. Packt Publishing.

Appendix A

Contribution table:

ID	Contribution	Tasks
up2102739	33%	Section 4, section 5, section 1
up2541668	33%	Section 1, Section 2,
up2047214	33%	Section 3, section 1